

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415462
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Shinohara; Masayuki et al.

Display switching device, vehicle display system and automobile

Abstract

A display switching device (1) includes: a lens array (40); a display part (10), comprising pixel regions (31) disposed to include regions in which light in which respective light emitted from light sources (50) is condensed by respective lenses (41) of the lens array (40) passes through. A transmittance of each pixel region (31) is set corresponding to a predetermined stationary pattern. Display of the display part (10) performed through irradiation of the light of the light sources (50) exhibits directivity, and a first averaged oriented direction (D1) averaging, among the light sources (50), oriented directions as directions of the directivity is inclined at a predetermined angle with respect to a normal direction of a display surface (11) of the display part (10).

Inventors: Shinohara; Masayuki (Kyoto, JP), Mori; Yuto (Kyoto, JP), Kurata; Gouo (Kyoto, JP), Mishina; Masao (Kyoto, JP), Okayasu; Yutaka (Kyoto, JP)

Applicant: OMRON Corporation (Kyoto, JP)

Family ID: 1000008821595

Assignee: OMRON Corporation (Kyoto, JP)

Appl. No.: 18/461498

Filed: September 05, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240131997 A1	Apr. 25, 2024

Foreign Application Priority Data

JP	2022-166388	Oct. 17, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: B60R11/02 (20060101); G02F1/1335 (20060101); G02F1/1343 (20060101)

U.S. Cl.:

CPC B60R11/0235 (20130101); B60R11/0264 (20130101); G02F1/133504 (20130101);
G02F1/133526 (20130101); G02F1/133607 (20210101); G02F1/134345 (20210101);

Field of Classification Search

CPC: B60R (11/0235); G02F (1/133607)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7755263	12/2009	Park	313/111	H01J 11/44
7821185	12/2009	Park	428/690	G02B 5/003
7995166	12/2010	Uehara	349/95	G02F 1/133526
8446355	12/2012	Uehara	345/204	H04N 13/398
9778472	12/2016	Wei	N/A	G02B 3/00
10036941	12/2017	Takahashi	N/A	B60K 35/232
11412596	12/2021	Shinohara	N/A	G09G 3/20
2004/0169831	12/2003	Uehara	348/E13.028	G02F 1/133526
2009/0052164	12/2008	Kashiwagi	362/97.2	H04N 13/32
2009/0086466	12/2008	Sugita	362/97.1	G02F 1/133615
2013/0242402	12/2012	Konig	359/630	G02B 27/0101
2014/0022619	12/2013	Woodgate	359/240	G02B 27/0093
2015/0029442	12/2014	Koike	349/65	G02B 5/045
2016/0053961	12/2015	Akutsu	362/521	F21S 43/14
2017/0343826	12/2016	De Zwart	N/A	G02B 30/27
2018/0329122	12/2017	Goto	N/A	G02B 27/0172
2023/0099347	12/2022	Matsumura	362/488	B60Q 3/62

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2003084681	12/2002	JP	N/A

Primary Examiner: Habib; Irfan

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Japan application serial No. 2022-166388, filed on Oct. 17, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a display switching device, a vehicle display system, and an automobile.

Description of Related Art

(3) As an example, Patent Document 1 discloses a vehicle-mounted display device having directivity in a specific direction. The vehicle-mounted display device is a liquid crystal display device including a liquid crystal panel. The vehicle-mounted display device exhibits directivity in the luminance distribution and the viewing angle by using a prism sheet that refracts light from a backlight as the light source of the liquid crystal panel and polarization plates provided in front of and behind the liquid crystal panel.

PRIOR ART DOCUMENT(S)

Patent Document(s)

(4) [Patent Document 1] Japanese Laid-open No. 2003-84681

(5) However, in the vehicle-mounted display device of Patent Document 1, the light from the backlight is emitted into the liquid crystal panel via a light guide plate, a light reflection member, a downward prism sheet, a diffusion sheet, an upward prism sheet, a light control film, and a polarization plate. That is, in the liquid crystal display, an image is displayed on the liquid crystal panel by using light of a light source traveling through multiple components. In addition, in a liquid crystal system, since only light of a specific polarization component is used for display, the light usage efficiency is not favorable. Therefore, the luminance of the liquid crystal panel is insufficient. In particular, under sunlight during daytime, it is difficult to see the image displayed on the liquid crystal panel.

(6) An aspect of the disclosure provides a display switching device having an increased usage efficiency of the light emitted from the light source, and exhibiting directivity in a display direction.

SUMMARY

(7) A display switching device according to an aspect of the disclosure switches display images by switching irradiation of light from a plurality of light sources. The display switching device includes: a lens array, in which a plurality of lenses are arranged; and a display part, including a plurality of pixel regions disposed to include regions in which light in which the respective light emitted from the light sources is condensed by the respective lenses of the lens array passes through. A transmittance of each of the pixel regions is set corresponding to a predetermined stationary pattern, and display of the display part performed through the irradiation of the light of the light sources exhibits directivity, and an averaged oriented direction averaging, among the light sources, oriented directions that are directions of the directivity is inclined at a predetermined angle with respect to a normal direction of a display surface of the display part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a shift lever including a display switching device.

(2) FIG. 2 is a schematic view illustrating a basic configuration of a display switching device according to Embodiment 1 of the disclosure.

(3) (a) and (b) of FIG. 3 are views illustrating an example of a configuration of a lens array shown in FIG. 2.

(4) FIG. 4 is a view illustrating an example of a state of light condensation by using the lens array.

(5) FIG. 5 is a view illustrating an example of switching display of a display part.

(6) FIG. 6 is a view illustrating a corresponding relationship among the display part, lenses forming the lens array and the respective light sources forming a first light source group.

- (7) FIG. 7 is a view illustrating a position relationship of the first light source group with respect to the display part and the lens array.
- (8) FIG. 8 is a view illustrating main parts of a configuration a display switching device according to a modified example of Embodiment 1.
- (9) (a) and (b) of FIG. 9 are views illustrating relationships between averaged oriented directions and users able to visually recognize an image in a case where the averaged oriented direction is arranged as two directions different from each other.
- (10) (a) to (e) of FIG. 10 are views illustrating display switching devices according to modified examples of Embodiment 1.
- (11) (a) and (b) of FIG. 11 are views illustrating a display switching device according to a modified example of Embodiment 1.

DESCRIPTION OF THE EMBODIMENTS

(12) A display switching device according to an aspect of the disclosure switches display images by switching irradiation of light from a plurality of light sources. The display switching device includes: a lens array, in which a plurality of lenses are arranged; and a display part, including a plurality of pixel regions disposed to include regions in which light in which the respective light emitted from the light sources is condensed by the respective lenses of the lens array passes through. A transmittance of each of the pixel regions is set corresponding to a predetermined stationary pattern, and display of the display part performed through the irradiation of the light of the light sources exhibits directivity, and an averaged oriented direction averaging, among the light sources, oriented directions that are directions of the directivity is inclined at a predetermined angle with respect to a normal direction of a display surface of the display part.

(13) According to the configuration, since display is performed through the light condensed by the respective lenses of the lens array, the display using the display part exhibits directivity of a certain extent. In addition, the averaged oriented direction averaging the oriented directions of the display of the display part performed through irradiation of the light from the respective light sources is inclined at the predetermined angle with respect to the normal direction of the display surface. That is, according to the configuration, display exhibiting directivity in a biased direction that is not the front surface with respect to the display surface of the display part can be realized. At this time, compared with the configuration using light of a specific polarization component, as in the conventional liquid crystal display device, for example, the usage efficiency of the light emitted from the light source can be higher. In addition, the cost can be lower than the liquid crystal display device.

(14) In the display switching device, it may also be that the lens located at a closest position with respect to a gravity center of the lens array is set as a reference lens, and a direction of a line segment connecting a main point of the reference lens and a gravity center position of a first light source group formed by a plurality of light sources of the light sources is inclined with respect to the normal direction of the display surface.

(15) According to the configuration, by shifting the gravity center position of the first light source group with respect to the gravity center position of the lens array, an angle can be set for the incident angle of the light from the first light source group with respect to the lens array. Therefore, the direction averaging the directions in which the light of the first light source group condensed by the respective lenses of the lens array passes through can be inclined with respect to the normal direction of the display surface of the display part. Accordingly, by disposing the pixel regions of the display part in the regions in which the light of the first light source group condensed by the respective lenses of the lens array passes through, display images having directivity in a first direction can be displayed on the display part.

(16) In the display switching device, it may also be that the display switching device further includes an optical member having a function of refracting light emitted from the display surface in a direction inclined with respect to the normal direction of the display surface, diffracting the light

emitted from the display surface in the direction inclined with respect to the normal direction of the display surface, or combining the refracting and the diffracting.

(17) According to the configuration, the light condensed by the respective lenses of the lens array is refracted and/or diffracted by the optical member, so as to be inclined with respect to the normal direction of the display surface. Accordingly, the display switching device having display directivity can be provided by using a simple configuration.

(18) In the display switching device, it may also be that the optical member is a lens.

(19) In the display switching device, it may also be that the optical member is a prism.

(20) In the display switching device, it may also be that the averaged oriented direction is arranged as two directions different from each other.

(21) According to the configuration, display exhibiting directivity in two directions can be realized. Accordingly, display that can be visually recognized at positions of two users can be carried out.

(22) In the display switching device, it may also be that display in accordance with a first averaged oriented direction is realized through light from a first light source group formed by a plurality of light sources of the light sources, and a second averaged oriented direction is realized through light from a second light source group formed by a plurality of light sources of the light sources and disposed at a position different from that of the first light source group.

(23) According to the configuration, the display switching device in which the first averaged oriented direction according to the light from the first light source group and the second averaged oriented direction according to the light from the second light source group are different can be realized.

(24) In the display switching device, it may also be that the display switching device further includes an optical member having a function of refracting the light emitted from the display surface in two directions different from each other, diffracting the light emitted from the display surface in two directions different from each other, or combining the refracting and the diffracting.

(25) According to the configuration, a display switching device in which the averaged oriented direction is arranged as two directions different from each other can be provided by using the optical member.

(26) IA vehicle display system according to another aspect of the disclosure is configured to include the display switching device according to any one of the above aspects, and the display switching device is provided in a state in which the averaged oriented direction is oriented toward an assumed seated position where a passenger is assumed to be seated in a vehicle.

(27) According to the configuration, a vehicle display system in which directivity is set in the display direction in accordance with the position of the passenger can be provided.

(28) In the vehicle display system, it may also be that the display switching device is provided in a shift lever provided in the vehicle.

(29) An automobile according to an aspect of the disclosure may be configured to be provided with the vehicle display system according to any of the above aspects.

(30) According to an aspect of the disclosure, a display switching device having an increased usage efficiency of the light emitted from a light source, and exhibiting directivity in a display direction can be provided.

Embodiment 1

(31) The display switching device of the disclosure is a display device having directivity in a display direction. In the embodiment, a display switching device **1** as an aspect of the display switching device of the disclosure is described.

1. Application Example

(32) Firstly, referring to FIG. **1**, an example of a scenario where the display switching device is applied is described. FIG. **1** is a view illustrating a shift lever **70** including the display switching device **1**. The display switching device **1** is provided in a vehicle display system **90** disposed in an automobile. In the embodiment, the vehicle display system **90** is disposed in the shift lever **70**

included in the automobile.

(33) As shown in FIG. 1, an image display part **72** is disposed on the upper surface of a grip part **71** of the shift lever **70**. An image in accordance with the operation on the shift lever **70** by the driver is displayed on the image display part **72**. On the image display part **72**, images respectively illustrating the respective transmission modes of the engine of the vehicle are displayed in accordance with the operation on the shift lever **70**. On the image display part **72**, letters indicating “P (Parking)”, “R (Reverse)”, “N (Neutral)”, and “D (Drive)” are displayed, for example.

(34) The image displayed on the image display part **72** exhibits display directivity toward an assumed seated position where a passenger (user) is assumed to be seated in the vehicle. According to the configuration, a vehicle display system in which directivity is set in the display direction in accordance with the position of the passenger can be provided. The image display part **72** is realized by using the display switching device described in detail in the following.

2. Configuration Example

(35) In the following, the configuration of the display switching device **1** according to the embodiment is described in detail based on FIGS. 2 to 7. In the following description, the configuration relating to the display switching device of the disclosure is described by defining a coordinate system of three directions, namely X (X1-X2) direction (left-right direction), Y (Y1-Y2) direction (upper-lower direction), and Z (Z1-Z2) direction (front-rear direction) as shown in FIG. 2, etc.

(36) (Basic Configuration of the Display Switching Device)

(37) FIG. 2 is a schematic view illustrating a basic configuration of the display switching device **1** according to the embodiment. As shown in FIG. 2, the display switching device **1** includes a display part **10**, and a microlens array (briefly referred to as lens array in the following) **40**. In addition, the display switching device **1** may further include a first light source group **51**, and a substrate **3** to which the first light source group **51** is installed.

(38) The display part **10** and the lens array **40** are supported by a housing **2**. In addition, in the embodiment, the basic configuration of the display switching device **1** is formed by installing the housing **2** to the substrate **3** to which the first light source group **51** is installed.

(39) The display switching device **1** may also include a protection layer for preventing damages above (Z1 side) of the display part **10**. In addition, the display switching device **1** may also include a light modulation member above the display part **10**. As the light modulation member, example may include a smoked member, a half mirror, a polarization plate, a color plate, a diffusion plate, a member displaying a display image, and a smoked member combined with a diffusion plate to provide a diffusion function, etc. In addition, a member blocking a portion of the viewing angle of the display part **10** may also be provided above the display part **10**. Examples of the member blocking a portion of the viewing angle include a louver, etc. The size of each of the above components may be designed as appropriate in accordance with the size of the image display part **72**.

(40) The first light source group **51** is formed by multiple light sources **50** emitting light toward the lens array **40**. The light sources **50** may be respectively configured to emit light of the same color, and may also be respectively configured to emit light of different colors. Examples of the light sources **50** include RGBLEDs.

(41) The configuration of the lens array **40** is described with reference to (a) and (b) of FIG. 3. (a) and (b) of FIG. 3 are views illustrating an example of a configuration of the lens array **40** shown in FIG. 2. A symbol **100** in (a) of FIG. 3 represents a view in which the lens array **40** shown in FIG. 2 is observed from Z1 direction. A symbol **101** in (a) of FIG. 3 represents a perspective view of the lens array **40** in which a portion of the view of the symbol **100** surrounded by a dot line is enlarged and shown. A view represented by a symbol **102** in (b) of FIG. 3 is a partial cross-sectional view in which a portion of the lens array **40** is enlarged and shown.

(42) As shown by the views represented by the symbols **100** to **102** in (a) and (b) of FIG. 3, the lens

array **40** is formed by arranging multiple lenses **41** into a two-dimensional matrix. The respective lenses **41** of the lens array **40** condense the light emitted from the light sources **50** of the first light source group **51**.

(43) FIG. **4** is a view illustrating an example of a state of light condensation by using the lens array **40**. According to FIG. **4**, it is understood that, by changing the position of the light source **50**, the light condensation location can be changed.

(44) As indicated by the symbols **100** and **101** in (a) of FIG. **3**, the center position of the region in which the lenses **41** are arranged into a two-dimensional matrix is a gravity center position **C1** of the lens array **40**. The following description is made by adopting the lens **41** located at a position closest to the gravity center position **C1** as a reference lens **41A**.

(45) The display part **10** is formed by an image layer **20** and a matrix layer **30**, and displays an image **P** (display image) to be displayed. The matrix layer **30** includes, for example, pixel regions **31** (opening, the same applies in the following) and pixel surrounding regions **32** (mask, the same applies in the following) as regions other than the pixel regions **31**. The image layer is bonded with the matrix layer **30**. Here, “pixel surrounding region **32**” refers to a region that is in the surrounding of each pixel region **31** and has a constant transmittance. In addition, the pixel surrounding region **32** shields the light from the side of the first light source group **51**, that is, shields the light from the side where the lens array **40** is disposed.

(46) The display part **10** includes multiple pixel regions **31** disposed to include regions in which the light in which the respective light emitted from the position of the first light source group **51** is condensed by the respective lenses **41** of the lens array **40** passes through. In the display part **10**, the transmittance of each of the pixel regions **31** is set corresponding to a predetermined stationary pattern.

(47) FIG. **5** is a view illustrating an example of switching display of the display part **10**. A view represented by a symbol **200** of FIG. **5** is a display example displayed on the display part (image layer **20**), a view represented by a symbol **201** is an example of a displayed pattern, a view represented by a symbol **202** is an enlarged view of a part **A** of the view represented by the symbol **200**, and a view represented by a symbol **203** is an enlarged view of a part **B** of the view represented by the symbol **202**.

(48) In the display example shown in FIG. **5**, for example, a display image **P1** (Japanese letter “ custom character” in the example), a display image **P2** (pattern “ Δ ” in the example), a display image **P3** (capital letter “G” in the example), and a display image **P4** (numeral symbol “6” in the example) can be switched and displayed.

(49) As shown in the view of the symbol **203** of FIG. **5**, as an example, it is possible to switch the display by partitioning the entire display part **10** into multiple regions (multiple pixels), so that at most four pixel regions **31** are included in one region. In addition, as shown in the view of the symbol **203** of FIG. **5**, a region other than the pixel regions **31** of the display part **10** becomes the pixel surrounding region **32**.

(50) FIG. **6** is a view illustrating a corresponding relationship among the display part **10**, the lenses **41** forming the lens array **40** and the respective light sources **50** forming the first light source group **51**. In FIG. **6**, two pixels are exemplified, and the light from the respective light sources **50A** to **50D** of the first light source group are respectively condensed by the lens array and emitted from the corresponding pixel regions **31**. Each pixel of the display part **10** is divided into the respective pixel regions **31** and the pixel surrounding region **32**. The light sources **50A** to **50D** shown in FIG. **6** may also be light sources respectively emitting white light, green light, red light, blue light, for example.

(51) By using such display switching device **1**, animation display or switchable display can be performed without using a liquid crystal display. Accordingly, a display switching device that is more cost-effective than a liquid crystal display can be realized.

(52) (Regarding Display Directivity)

(53) The directivity of the display images displayed on the display part **10** of the display switching device **1** is described with reference to FIG. 7. FIG. 7 is a view illustrating a position relationship of the first light source group **51** with respect to the display part **10** and the lens array **40**. The display switching device **1** can perform display with directivity on the display part **10** by shifting the gravity center position C2 of the first light source group **51** from the center of the display part **10**.

(54) As shown in FIG. 7, each light source **50** of the first light source group **51** is disposed on the X2 side with respect to a reference line BL. Here, the reference line BL is the normal direction of a display surface **11** of the display part **10** of the display switching device **1**, and is a line passing through the center position of the display part **10**. In the embodiment, the reference line BL is a line passing through the gravity center position C1 of the lens array **40** and a main point S1 of the reference lens **41A**.

(55) The gravity center position C2 of the first light source group **51** is shifted in X2 direction with respect to the gravity center position C1 of the lens array **40**. Here, the gravity center position C2 of the first light source group **51** is a position corresponding to the middle position of a diagonal connecting the light sources **50** located on the outermost sides among the light sources **50**. The gravity center position C2 is a hypothetical position. The direction of a line segment L1 connecting the main point S1 of the reference lens **41A** of the lens array **40** and the gravity center position C2 of the first light source group **51** is inclined with respect to the reference line BL. In other words, the line segment L1 is inclined with respect to the normal direction of the display surface **11**.

(56) The light from each light source **50** of the first light source group **51** is emitted at an angle with respect to the lens array **40**. The light of the first light source group **51** condensed by the lens array **40** is inclined at a predetermined angle with respect to the normal direction of the display surface **11** of the display part **10** and emitted to the display part **10**. The light emitted from the display part **10** is emitted in a state of being inclined at a predetermined angle with respect to the normal direction of the display surface **11**. More specifically, a first averaged oriented direction D1 averaging, among the light sources **50**, the oriented directions that are the directions of the directivity of the display of the display part **10**, is inclined at a predetermined angle α with respect to the normal direction of the display surface **11**. That is, the display in accordance with the first averaged oriented direction D1 is realized by using the light from the first light source group **51**. Here, the predetermined angle α may be 10 degrees or more. In addition, the predetermined angle α may be 15 degrees or more.

(57) That is, the display of the display part **10** through the irradiation of the light from the respective light sources **50** of the first light source group **51** exhibits directivity. When viewing in a direction at the predetermined angle with respect to the normal direction of the display surface **11** of the display part **10**, that is, viewing in a direction opposite the first averaged oriented direction D1, the user can visually recognize the image displayed on the display part **10**. However, when viewing in a direction at an angle other than the predetermined angle with respect to the normal direction of the display surface **11**, that is, viewing in a direction other than the direction opposite to the first averaged oriented direction D1, the user cannot visually recognize the image displayed on the display part **10**.

(58) According to the display switching device **1**, since the display is performed through the light condensed by the respective lenses **41** of the lens array **40**, the display using the display part exhibits directivity of a certain extent. In addition, the averaged oriented direction averaging the oriented directions of the display of the display part **10** performed through the irradiation of the light from the respective light sources **50** is inclined at the predetermined angle with respect to the normal direction of the display surface. That is, according to the configuration, display exhibiting directivity in a biased direction that is not the front surface with respect to the display surface **11** of the display part **10** can be realized. At this time, compared with the configuration using light of a specific polarization component, as in the conventional liquid crystal display device, for example,

the usage efficiency of the light emitted from the light source can be higher. In addition, the cost can be lower than the liquid crystal display device.

(59) In addition, according to the display switching device **1**, by shifting the gravity center position **C2** of the first light source group **51** with respect to the gravity center position **C1** of the lens array **40**, an angle can be set for the incident angle of the light from the first light source group **51** with respect to the lens array **40**. Therefore, the direction averaging the directions in which the light of the first light source group **51** condensed by the respective lenses **41** of the lens array **40** passes through can be inclined with respect to the normal direction of the display surface **11** of the display part **10**. Accordingly, by disposing the pixel regions **31** of the display part **10** in the regions in which the light of the first light source group **51** condensed by the respective lenses of the lens array passes through, display images having directivity can be displayed on the display part **10**.

(60) In addition, according to the display switching device **1**, the light usage efficiency is increased. As a result, even if the first averaged oriented direction **D1** is inclined by 10 degrees or more with respect to the normal direction of the display surface **11**, the visibility of the display in the display part **10** can be increased. In addition, even if the first averaged oriented direction **D1** is inclined by 15 degrees or more with respect to the normal direction of the display surface **11**, in the vehicle display system **90** included in the shift lever **70** of the automobile, for example, the visibility of the display in the display part **10** can be increased.

3. Modified Example

(61) Although the embodiments of the disclosure have been described above in detail, the above description is merely an example of the disclosure in every respect. It goes without saying that various modifications and variations can be made without departing from the scope of the disclosure. For example, the following changes are possible. In the following, the same symbols are used for components similar to those in the embodiment, and the description is omitted as appropriate for points similar to the embodiment. The following modified examples can be combined as appropriate.

(62) <3.1>

(63) In the display switching device **1** in Embodiment 1, a configuration including only the first light source group **51** is described. However, the display switching device of the disclosure is not limited thereto.

(64) A display switching device **1A** according to an aspect of the disclosure is described with reference to FIG. **8** and (a) and (b) of FIG. **9**. FIG. **8** is a view illustrating main parts of a configuration of the display switching device **1A** according to a modified example of Embodiment 1. In the display switching device **1A**, a second light source group **52** formed by multiple light sources **50B** is further provided. In FIG. **8** and (a) and (b) of FIG. **9**, the light sources **50** forming the first light source group **51** are set as light sources **50A**, and the light sources **50** forming the second light source group **52** are set as light sources **50B**.

(65) As shown in FIG. **8**, the second light source group **52** is arranged at a position different from the first light source group **51**. More specifically, the second light source group **52** is disposed at the **X1** side with respect to the reference line **BL**. A gravity center position **C3** of the second light source group **52** is shifted in **X1** direction with respect to the gravity center position **C1** of the lens array **40**. The direction of a line segment **L2** connecting the main point **S1** of the reference lens **41A** of the lens array **40** and the gravity center position **C3** of the second light source group **52** is inclined with respect to the reference line **BL**. In other words, the line segment **L2** is inclined with respect to the normal direction of the display surface **11**. It is noted that the line segment **L2** may be inclined with respect to the normal direction of the display surface **11** at an inclined angle same as or different from that of the line segment **L1** related to the first light source group **51**. In addition, the line segment **L2** may be in the same direction as the normal direction of the display surface **11**.

(66) The light from each light source **50** of the second light source group **52** is emitted at an angle with respect to the lens array **40**. The light of the second light source group **52** condensed by the

lens array **40** is inclined at a predetermined angle with respect to the normal direction of the display surface **11** of the display part **10** and emitted to the display part **10**. The light of the second light source group **52** emitted from the display part **10** is emitted in a state of being inclined at a predetermined angle with respect to the normal direction of the display surface **11**. More specifically, a second averaged oriented direction **D2** averaging, among the light sources **50B**, the oriented directions that are the directions of the directivity of the display part **10** is inclined at a predetermined angle $f3$ with respect to the normal direction of the display surface **11**. That is, the second averaged oriented direction **D2** is realized by using the light from the second light source group **52**. The second averaged oriented direction **D2** may also be in the same direction as the normal direction of the display surface **11**.

(67) According to the configuration, the first averaged oriented direction **D1** of the first light source group **51** and the second averaged oriented direction **D2** of the second light source group **52** are different directions. Therefore, display exhibiting directivity in two directions can be realized. Accordingly, as shown in (a) and (b) of FIG. **9**, display that can be visually recognized at positions of two users can be carried out. In addition, the display switching device **1** in which the first averaged oriented direction **D1** according to the light from the first light source group **51** and the second averaged oriented direction **D2** according to the light from the second light source group **52** are different can be realized.

(68) (a) and (b) of FIG. **9** are views illustrating relationships between the averaged oriented directions and users able to visually recognize the image in a case where the averaged oriented direction is arranged as two directions different from each other. A symbol **300** in (a) of FIG. **9** represents a diagram illustrating a case where the first light source group **51** and the second light source group **52** are separated in the left-right direction. A symbol **301** in (b) of FIG. **9** represents a diagram illustrating a case where the first light source group **51** and the second light source group **52** are separated in the upper-lower direction. In (a) and (b) of FIG. **9**, with respect to the display surface **11** of the display part **10**, the description is made by setting X direction as the left-right direction, Y direction as the upper-lower direction, and Z direction as the depth direction.

(69) As indicated with the diagram of the symbol **300** in (a) of FIG. **9**, in the case where the first light source group **51** and the second light source group **52** are disposed at different positions in the left-right direction, display that is visible at two different positions for two users present at different positions is carried out. A user **U1** present at a position opposite to the first averaged oriented direction **D1** cannot visually recognize the display image displayed through the second light source group **52**, but can visually recognize the display image displayed through the first light source group **51**. Meanwhile, a user **U2** present at a position opposite to the second averaged oriented direction **D2** cannot visually recognize the display image displayed through the first light source group **51**, but can visually recognize the display image displayed through the second light source group **52**. According to such configuration, for example, it is possible to reinforce the directivity of display at two positions, which are the user seated position on the driver's side and the user seated position on the passenger's side, for example.

(70) As indicated with the symbol **301** in (b) of FIG. **9**, in the case where the first light source group **51** and the second light source group **52** are disposed at different positions in the upper-lower direction, display that is visible at two different positions are possible for one user. For example, in the case where the second averaged oriented direction **D2** of the second light source group **52** is set according to an assumed seated position where the user **U1** is assumed to be seated in the vehicle, the display image of the display part **10** viewed by the user **U1** from the assumed seated position with a normal sight line is the image displayed through the second light source group **52**.

Meanwhile, when the user **U1** moves the sight line downward (**Y2** direction) from the normal sight line, since the sight line of the user **U1** is opposite to the first averaged oriented direction **D1** of the first light source group **51**, the user **U1** can visually recognize the image displayed by the first light source group **51**.

(71) <3.2>

(72) In Embodiment 1, the display directivity in the display part **10** of the display switching device **1** is made possible by shifting the gravity center position **C2** of the first light source group **51** from the center of the display part **10**. However, the display switching device according to an aspect of the disclosure may be configured to be able to perform display exhibiting directivity in the display part **10** by including an optical member that refracts light. The display switching device according to the modified example includes a lens as an optical member **60** that refracts the light emitted from the display surface **11** of the display part **10** in a direction inclined with respect to the normal direction of the display surface **11**.

(73) The display switching device according to the modified example is described with reference to (a) to (e) of FIG. **10**. (a) to (e) of FIG. **10** are views illustrating display switching devices according to modified examples of Embodiment 1.

(74) A symbol **400** in (a) of FIG. **10** represents a view illustrating main parts of the configuration of a display switching device **1B** including a convex lens **61** as the optical member **60**. A symbol **401** in (b) of FIG. **10** represents a diagram illustrating the simulation result of the display directivity by using the display part **10** of the display switching device **1B**.

(75) As indicated with the view of the symbol **400** in (a) of FIG. **10**, the display switching device **1B** includes the convex lens **61** that is the optical member **60** on the light emitting side (**Z1** side) of the display part **10**. The convex lens **61** converges the light of the first light source group **51** emitted from the display part **10**.

(76) The display directivity according to the display switching device **1B** is described with reference to the simulation result shown in the diagram of the symbol **401** in (b) of FIG. **10**. Through observing the light intensity distribution shown on the upper left in the diagram of the symbol **401**, it is known that the portion where the light intensity is high in the left-right direction of the diagram is split into two. In the graph indicating light intensity as shown below the diagram of the light intensity distribution of the symbol **401**, it is known that there are two peaks at about ± 10 degrees.

(77) That is, the display switching device **1B** including the convex lens **61** exhibits two averaged oriented directions at about ± 10 degrees. The convex lens **61** refracts the light emitted from the display surface **11** in two directions different from each other. Accordingly, it is possible to perform display having directivity in the display part **10** by using the display switching device **1B** including the convex lens **61**.

(78) A symbol **410** in (c) of FIG. **10** represents a view illustrating main parts of the configuration of a display switching device **1C** including a concave lens **62** as the optical member **60**. A symbol **411** in (d) of FIG. **10** represents a diagram illustrating the simulation result of the display directivity by using the display part **10** of the display switching device **1C**.

(79) As indicated with the diagram of the symbol **410** in (c) of FIG. **10**, the display switching device **1C** includes the concave lens **62** that is the optical member **60** on the **Z1** side of the display part **10**. The concave lens **62** diverges the light of the first light source group **51** emitted from the display part **10**.

(80) The display directivity according to the display switching device **1C** is described with reference to the simulation result shown in the diagram of the symbol **411** in (d) of FIG. **10**. Through observing the light intensity distribution shown on the upper left in the diagram of the symbol **411**, it is known that the portion where the light intensity is high is split into two in the left-right direction of the diagram. In the graph indicating light intensity as shown below the diagram of the light intensity distribution of the symbol **411**, it is known that there are two peaks at about ± 15 degrees.

(81) That is, the display switching device **1C** including the concave lens **62** exhibits two averaged oriented directions at about ± 15 degrees. The concave lens **62** refracts the light emitted from the display surface **11** in two directions different from each other. Accordingly, it is possible to perform

display having directivity in the display part **10** by using the display switching device **1C** including the concave lens **62**.

(82) A symbol **420** in (e) of FIG. **10** represents a view illustrating main parts of the configuration of a display switching device **1D** including a lens **63** as the optical member **60**. As indicated with the view of the symbol **420**, the display switching device **1D** may include the lens **63** in which a convex lens and a concave lens are combined as the optical member **60**. The lens **63** is provided on the **Z1** side of the display part **10**.

(83) According to the types of the lenses **61** to **63** of the display switching devices **1B** to **1D**, the display image displayed on the display part **10** may be scaled up or down. Therefore, by appropriately choosing a convex lens or a concave lens in accordance with the size of the display image displayed on the display part **10**, display with variation in image size can be performed. Accordingly, a display switching device capable of performing display that is easy to draw the user's attention can be provided.

(84) The lenses **61** to **63** of the display switching devices **1B** to **1D** are not limited to the configurations shown in (a) to (e) of FIG. **10**. The shapes of the lenses **61** to **63** may also be asymmetric. In the case where the directivity is only reinforced for a specific user, the shapes of the lenses **61** to **63** may be asymmetric. For example, in a case where the display switching devices **1B** to **1D** are applied to a button on the driver's side of an automobile, it is necessary to arrange the directivity toward the driver seated in front of the button and the passenger seated at the passenger seat. Therefore, the shapes of the lenses **61** to **63** may be left-right asymmetric. In addition, a protection layer for preventing damage or a reflection prevention layer that prevents light reflection may be formed on the surfaces of the lenses **61** to **63** of the display switching devices **1B** to **1D**.

(85) It is noted that the position where the first light source group **51** is disposed in (a) to (e) of FIG. **10** is not limited to the examples shown in the figure, but can be changed as appropriate in accordance with the averaged oriented direction of display that is set (the same applies to (a) and (b) of FIG. **11**). In addition, the lenses **61** to **63** of the display switching devices **1B** to **1D** may also be provided on the **Z2** side of the display part **10**. A prism may be incorporated into the lenses **61** to **63** of the display switching devices **1B** to **1D**, and a member that diffracts light may also be incorporated into the lenses **61** to **63** of the display switching devices **1B** to **1D**. In addition, the display switching device may also include an optical member that diffracts light in place of the optical member that refracts light.

(86) <3.3>

(87) A display switching device **1E** according to an aspect of the disclosure may be configured to able to perform display exhibiting directivity in the display part **10** by including an optical member that refracts light. The display switching device **1E** according to the modified example includes a prism as the optical member **60** that refracts the light emitted from the display surface **11** of the display part **10** in a direction inclined with respect to the normal direction of the display surface **11**.

(88) The display switching device **1E** is described with reference to (a) and (b) of FIG. **11**. (a) and (b) of FIG. **11** are views illustrating the display switching device **1E** according to a modified example of Embodiment 1. A symbol **500** in (a) of FIG. **11** represents a view illustrating main parts of the configuration of the display switching device **1E** including a prism **64** as the optical member **60**. A symbol **501** in (a) of FIG. **11** represents a view in which a portion surrounded by a circle in the view of the symbol **500** is enlarged. A symbol **502** in (b) of FIG. **11** represents a diagram illustrating the simulation result of the display directivity by using the display part **10** of the display switching device **1E**.

(89) As indicated with the view of the symbol **500** in (a) of FIG. **11**, the display switching device **1E** includes multiple prisms **64** that are the optical members **60** on the **Z1** side of the display part **10**. The substrates of the prisms **64** are disposed to be located on the **Z2** side. In other words, the substrates of the prisms **64** are disposed to be opposite to the pixel regions **31** of the display part **10**.

(90) As indicated with the symbol **501** in (a) of FIG. **11**, the light of the first light source group **51** is emitted from the pixel regions **31** of the display part **10** into the substrate of each prism **64**. The light emitted into the substrate of each prism **64** is refracted and emitted in the normal direction of the surface **64A** and the normal direction of the surface **64B** of each prism **64**.

(91) The display directivity according to the display switching device **1E** is described with reference to the simulation result shown in the diagram of the symbol **502** in (b) of FIG. **11**. Through observing the light intensity distribution shown on the upper part in the diagram of the symbol **502**, it is known that the portion where the light intensity is high is split into two in the left-right direction of the diagram. In the graph indicating light intensity as shown below the diagram of the light intensity distribution of the symbol **502**, it is known that there are two peaks at about ± 20 degrees.

(92) That is, the display switching device **1E** including the prisms **64** exhibits two averaged oriented directions at about ± 20 degrees. Each prism **64** refracts the light emitted from the display surface **11** in two directions different from each other. Accordingly, it is possible to perform display having directivity in the display part **10** by using the display switching device **1E** including the prisms **64**.

(93) In addition, each of the prisms **64** of the display switching device **1E** may also be provided on the **Z2** side of the display part **10**. In addition, the display switching device **1E** may also have a configuration including one prism **64**. Also, the substrate of the prism **64** may also be located on the **Z1** side. In addition, the prism **64** may be in a right-angled triangle shape, and may also be configured as having a surface parallel with the normal direction of the display surface **11** of the display part **10** on the side of **Z1**. According to such configuration, the display of the display part **10** can exhibit one averaged oriented direction. In addition, the angles of the inclined surfaces of the prism **64** may also be set to be different angles of the inclined surfaces in accordance with the portion of the prism **64** where the light of the first light source group **51** is emitted. According to such configuration, the angle of display directivity is further controlled, and the display image visually recognized by the user can be clearer.

(94) In addition, the prism may also be configured to be incorporated into the lens array **40**. According to such configuration, the accumulation of dirt such as dust in the gaps (microstructures) between the prisms **64** can be reduced. In addition, through integral formation, the cost of the display switching device can be reduced.

(95) According to the display switching devices **1B** to **1E**, the light condensed by the respective lenses **41** of the lens array **40** is refracted and/or diffracted by the optical member **60**, so as to be inclined with respect to the normal direction of the display surface **11** of the display part **10**. Accordingly, the display switching device having display directivity can be provided by using a simple configuration.

(96) In addition, according to the display switching devices **1B** to **1E**, a display switching device in which the averaged oriented direction is arranged as two directions different from each other can be provided by using the optical member **60**.

4. Application Example

(97) In the embodiments having been described so far, the display switching device of the disclosure is applied to a shift lever. However, the display switching device of the disclosure can also be applied to other driving devices of the vehicle.

(98) For example, the display switching device of the disclosure can be applied to a wiper switch controlling the operation mode of a wiper.

(99) In addition, the display switching device of the disclosure can also be applied to a dial-type system selecting the operation mode of a transmission.

(100) In addition, the display switching device of the disclosure can be applied to a switch (also referred to as a steering switch) provided on a steering wheel.

(101) In addition, the display switching device of the disclosure can also be applied in an operation

panel for the back seat. Such operation panel may be, for example, capable of controlling the air conditioning of the vehicle and selecting the audio operation. Accordingly, the passenger at the back seat can control some of the driving devices of the vehicle.

(102) In addition, the display switching device of the disclosure can be applied to a so-called push start switch that controls ON/OFF of the power of the engine. In such case, the color of the image displayed in the case where the power of the engine is OFF and the driver applies brakes of the vehicle and the case where the power of the engine is OFF and the driver does not apply brakes of the vehicle may be changed.

(103) In addition, the display switching device of the disclosure can be applied to an instrument panel. For example, the display switching device of the disclosure may be applied to a turn signal or alert display displayed on the instrument panel.

(104) In addition, the display switching device of the disclosure can be applied to a light switch that controls the operation of the headlight of the vehicle.

(105) In addition, the display switching device of the disclosure can be applied to a heater control switch that controls the operation of a heater of the vehicle.

(106) In addition, the display switching device of the disclosure can be applied to a welfare vehicle. Here, as a welfare vehicle, a welfare vehicle transporting a wheelchair in which a person sits is known. In such welfare vehicle, various switches are present, such as a switch for moving the wheelchair into the vehicle, such as a height adjustment switch, a safety belt switch, a fixing device for fixing the wheelchair. In addition, depending on the switches, multiple operations, such as “pull”, “push”, and “long press”, are required. Therefore, there is an issue that it is difficult to verify how to operate each switch of the welfare vehicle. By applying the display switching device of the disclosure to each switch of such welfare vehicle, it becomes easier for the operator of the welfare vehicle to recognize which mode is the currently selected mode, and the concern of generating an erroneous operation can be alleviated.

(107) In addition, the display switching device of the disclosure is not limited to being applied to a vehicle-related device. The display switching device of the disclosure may also be applied to a device or a system which is a device or a system provided in a scenario other than a vehicle and has the needs to perform display with directivity.

(108) In the application examples of the display switching device, in the case where the display switching device is applied as a switch, by providing a dent at the central portion of the display switching device, the touchability of the user can be facilitated, while the display has directivity.

(109) In the application examples of the display switching device, in the case where the display switching device is applied to a grip portion, such as a shift lever, by making the display surface side of the display switching device a convex shape, the grip portion can become easier to grip. Accordingly, the operability can be facilitated, while the display has directivity.

(110) The disclosure is not limited to the above-described embodiments, and various modifications are possible within the scope of the claims. Embodiments obtained by appropriately combining technical means disclosed in different embodiments are also included in the technical scope of the disclosure.

(111) [Outline]

(112) A display switching device according to Aspect 1 of the disclosure switches display images by switching irradiation of light from a plurality of light sources. The display switching device includes: a lens array, in which a plurality of lenses are arranged; and a display part, including a plurality of pixel regions disposed to include regions in which light in which the respective light emitted from the light sources is condensed by the respective lenses of the lens array passes through. A transmittance of each of the pixel regions is set corresponding to a predetermined stationary pattern, and display of the display part performed through the irradiation of the light of the light sources exhibits directivity, and an averaged oriented direction averaging, among the light sources, oriented directions that are directions of the directivity is inclined at a predetermined angle

with respect to a normal direction of a display surface of the display part.

(113) According to a display switching device of Aspect 2 of the disclosure, in Aspect 1, it may also be that the lens located at a closest position with respect to a gravity center of the lens array is set as a reference lens, and a direction of a line segment connecting a main point of the reference lens and a gravity center position of a first light source group formed by a plurality of light sources of the light sources is inclined with respect to the normal direction of the display surface.

(114) According to a display switching device of Aspect 3 of the disclosure, in Aspect 1 or 2, it may also be that the display switching device further includes an optical member having a function of refracting light emitted from the display surface in a direction inclined with respect to the normal direction of the display surface, diffracting the light emitted from the display surface in the direction inclined with respect to the normal direction of the display surface, or combining the refracting and the diffracting.

(115) According to a display switching device of Aspect 4 of the disclosure, in Aspect 3, it may also be that the optical member is a lens.

(116) According to a display switching device of Aspect 5 of the disclosure, in Aspect 3, it may also be that the optical member is a prism.

(117) According to a display switching device of Aspect 6 of the disclosure, in any one of Aspects 1 to 5, it may also be that the averaged oriented direction is arranged as two directions different from each other.

(118) According to a display switching device of Aspect 7 of the disclosure, in Aspect 6, it may also be that display in accordance with a first averaged oriented direction is realized through light from a first light source group formed by a plurality of light sources of the light sources, and a second averaged oriented direction is realized through light from a second light source group formed by a plurality of light sources of the light sources and disposed at a position different from that of the first light source group.

(119) According to a display switching device of Aspect 8 of the disclosure, in Aspect 6 or 7, it may also be that the display switching device further includes an optical member having a function of refracting the light emitted from the display surface in two directions different from each other, diffracting the light emitted from the display surface in two directions different from each other, or combining the refracting and the diffracting.

(120) A vehicle display system according to Aspect 9 of the disclosure is configured to include the display switching device according to any one of Aspects 1 to 8, and the display switching device is provided in a state in which the averaged oriented direction is oriented toward an assumed seated position where a passenger is assumed to be seated in a vehicle.

(121) According to a vehicle display system of Aspect 10 of the disclosure, in Aspect 9, it may also be that the display switching device is provided in a shift lever provided in the vehicle.

(122) An automobile according to Aspect 11 of the disclosure is configured to be provided with the vehicle display system according to Aspect 9 or 10.

Claims

1. A display switching device, switching display images by switching irradiation of light from a plurality of light sources, the display switching device comprising: a lens array, in which a plurality of lenses are arranged; and a display part, comprising a plurality of pixel regions disposed to comprise regions in which light in which the respective light emitted from the light sources is condensed by the respective lenses of the lens array passes through, wherein a transmittance of each of the pixel regions is set corresponding to a predetermined stationary pattern, and display of the display part performed through the irradiation of the light of the light sources exhibits directivity, and an averaged oriented direction averaging, among the light sources, oriented directions that are directions of the directivity is inclined at a predetermined angle with respect to a

normal direction of a display surface of the display part, wherein the lens located at a closest position with respect to a gravity center of the lens array is set as a reference lens, and a direction of a line segment connecting a main point of the reference lens and a gravity center position of a first light source group formed by a plurality of light sources of the light sources is inclined with respect to the normal direction of the display surface.

2. The display switching device as claimed in claim 1, further comprising an optical member having a function of refracting light emitted from the display surface in a direction inclined with respect to the normal direction of the display surface, diffracting the light emitted from the display surface in the direction inclined with respect to the normal direction of the display surface, or combining the refracting and the diffracting.

3. The display switching device as claimed in claim 2, wherein the optical member is a lens.

4. The display switching device as claimed in claim 2, wherein the optical member is a prism.

5. A vehicle display system, comprising the display switching device as claimed in claim 1, wherein the display switching device is provided in a state in which the averaged oriented direction is oriented toward an assumed seated position where a passenger is assumed to be seated in a vehicle.

6. The vehicle display system as claimed in claim 5, wherein the display switching device is provided in a shift lever provided in the vehicle.

7. An automobile, provided with the vehicle display system as claimed in claim 5.

8. A display switching device, switching display images by switching irradiation of light from a plurality of light sources, the display switching device comprising: a lens array, in which a plurality of lenses are arranged; and a display part, comprising a plurality of pixel regions disposed to comprise regions in which light in which the respective light emitted from the light sources is condensed by the respective lenses of the lens array passes through, wherein a transmittance of each of the pixel regions is set corresponding to a predetermined stationary pattern, and display of the display part performed through the irradiation of the light of the light sources exhibits directivity, and an averaged oriented direction averaging, among the light sources, oriented directions that are directions of the directivity is inclined at a predetermined angle with respect to a normal direction of a display surface of the display part, wherein the averaged oriented direction is arranged as two directions different from each other, wherein display in accordance with a first averaged oriented direction is realized through light from a first light source group formed by a plurality of light sources of the light sources, and a second averaged oriented direction is realized through light from a second light source group formed by a plurality of light sources of the light sources and disposed at a position different from that of the first light source group.

9. The display switching device as claimed in claim 8, further comprising an optical member having a function of refracting the light emitted from the display surface in two directions different from each other, diffracting the light emitted from the display surface in two directions different from each other, or combining the refracting and the diffracting.
